CEITEC

MUNI



C9555: A survival guide to simulations

Week 3

Denys Biriukov – Oct. 12th, 2023

More a discussion than monologue

If you have a quick question to catch up, raise your hand and ask – we will try to fix it and bring you back

If quick answer does not help, let's chat during a break or after

This course is heavily practical – theory will be trying to catch up, not vice versa

Get ready for practical session

- Wolf computers with GPU (more comfortable to work on)
  - wolf04 (not really good)
  - wolf05
  - wolf07
  - wolf08
  - wolf09
  - wolf12
  - wolf18
  - wolf19

Get ready for practical session

- A bit of things to be done:
  - `mv ~/survival_guide_simulations_c9555/week02/ ~/C9555/`
  - `rm -rf {SURVIVAL_GUIDE_DIRECTORY}`
    - **BE SURE THAT PREVIOUS COMMAND (MV) WAS SUCCESSFUL BEFORE DELETING**

- https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555

Get ready for practical session

- `git clone https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555.git`
- `cd survival_guide_simulations_c9555/`
- `cd week03/`
- `bash ../infinity/infinity-temp.sh`

- `bash ../conda/conda-adjust.sh`
  - `bash ../conda/conda-light-activate.sh` (if does not work, first do this)
- `bash ../conda/jupyter-start.sh`
- `bash ../conda/conda-adjust-mda.sh`

What is today?

- Week 3:
  Mixing Solvent – Physics and Chemistry in a Box.
- We prepared and run simulations with water/methanol mixtures.
- Today we will analyze them.
- Related theoretical concepts (touched the last week and today) will be discussed.

TEACHING WEEK	DATE	LECTURER	TOPIC
01	21.09	TR	Getting Started (Linux, Python, VMD)
02	05.10	DB	Mixing Solvents – Physics and Chemistry in a Box – part I
03	12.10	DB	Mixing Solvents – Physics and Chemistry in a Box – part II
04	19.10	DB	Ions, Molecules, and Their Interactions
05	26.10	OK	Making Drugs *in silico* – part I – Lipid Membranes
06	02.11	OK	Making Drugs *in silico* – part II – Ligands Parametrizations
07	09.11	TR	Making Drugs *in silico* – part III – Steered Molecular Dynamics & Free Energy Calculations
08	16.11	TR	Making Drugs *in silico* – part IV – Free Energy Calculations & Permeability Coefficient
09	23.11	WM	How to Beat AlphaFold – part I
10	30.11	WM	How to Beat AlphaFold – part II
11	07.12	WM	Working in a Professional Environment
12	14.12	OK	Zápočet – first session to start your project

Today program

Lecture (1st hour)

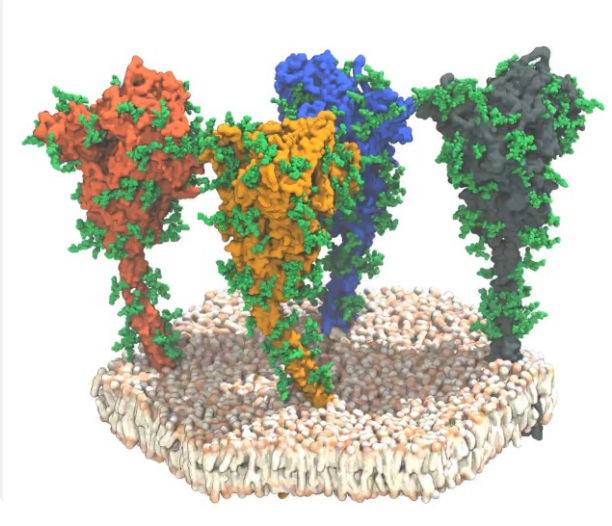
- Relation of molecular dynamics to thermodynamics.
- Ensembles in molecular simulations.
- Bridging simulation outputs and experimental observables.

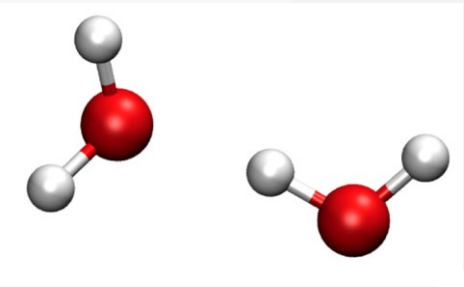
Practical session (2nd and 3rd hours)

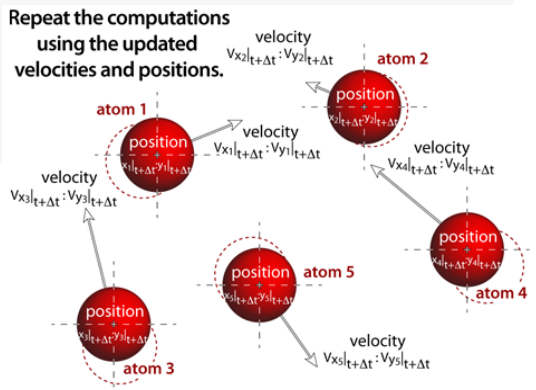
- Analysis of systems simulated in GROMACS.
- Convergence of simulation data and concepts of error estimation.
- Introduction to MDAnalysis as automatized and advanced post-processing.

Quick recall of the previous week

- Molecular dynamics simulations are based on physical laws and concepts.
- Atoms are represented as billiard balls that move according to Newton's equations of motion.
  - Born–Oppenheimer approximation.
- Force fields (predefined parameters) drive the evolution of the system.
- Time in simulation – time step, trajectories, equilibration, non-equilibrium systems.

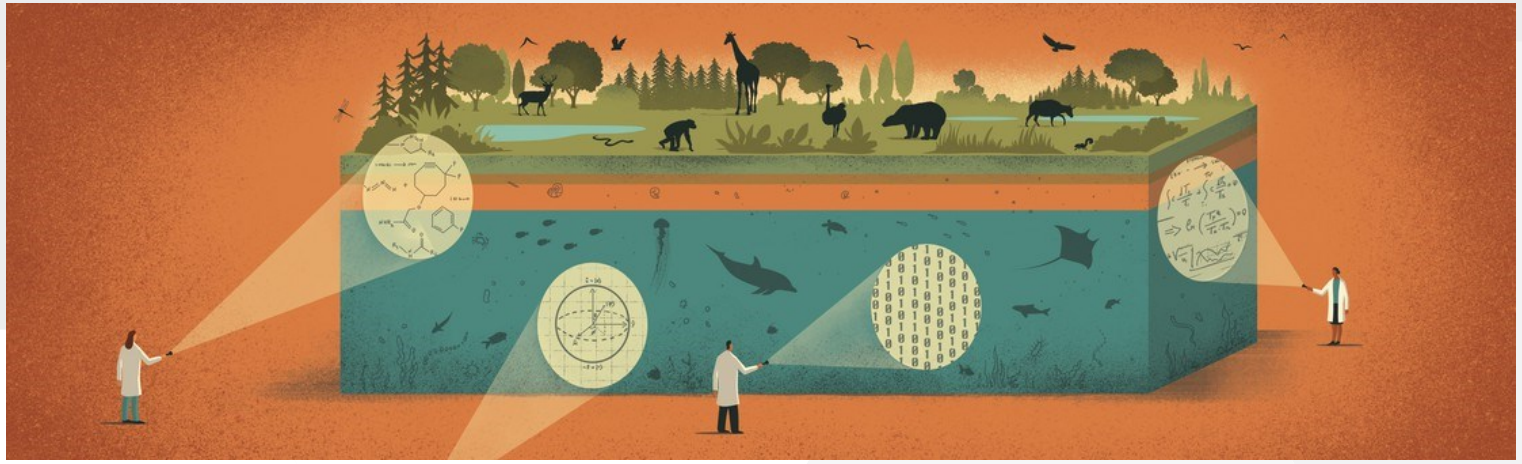






Bridging two worlds

- Simulations are a tool.
- We collect numbers.
- … but it is not the endgame.

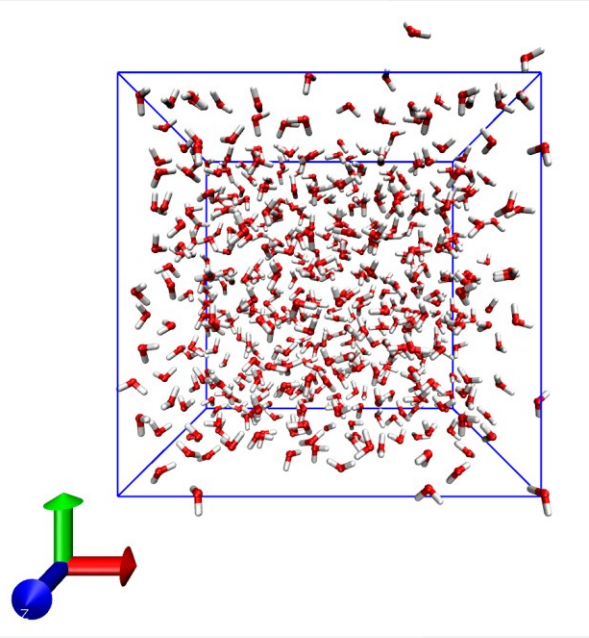


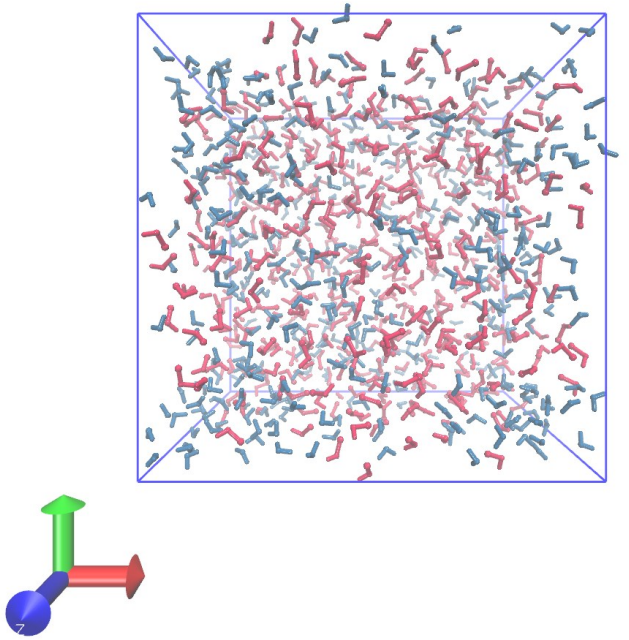
from https://elifesciences.org/subjects/physics-living-systems

- Massive amounts of numerical information should be transformed into answers to biological questions.
  - Simulation outputs compared to experimental observables.
- How to make this transformation?

Micro and macro

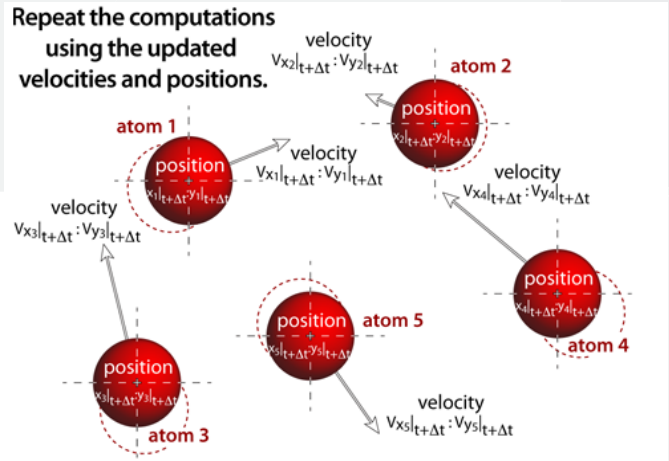
- Newton’s equations of motion govern microscopic evolution (atomic positions, atomic velocities).
- They do not explicitly give macroscopic properties (e.g., temperature, pressure, etc.)
- Macroscopic properties are obtained from trajectories via statistical mechanics (statistical thermodynamics).



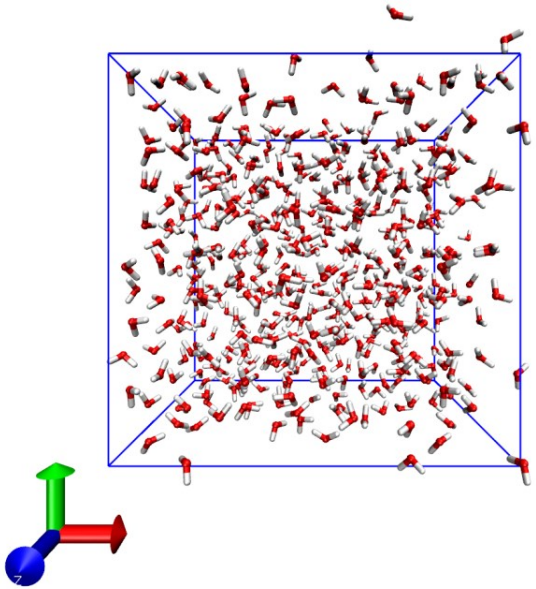


Micro and macro

- Microscopic properties: $r\ (x, y, z)$, $v\ (v_x, v_y, v_z)$



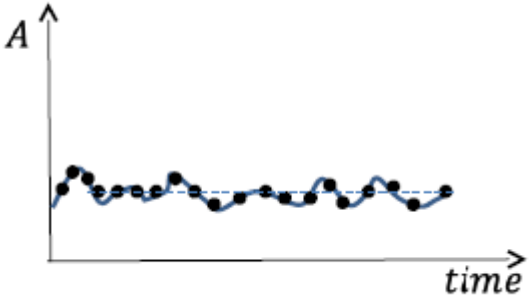
- Macroscopic properties: $T, P, V, \rho, \eta, \ldots$





Ensembles and time

- Ergodic hypothesis: all accessible microstates are equiprobable over a long period of time.
  - Average over ensemble – average over time.
- Macroscopic properties are given by probability distribution within an ensemble.
- In molecular dynamics simulations, we “move” within the ensemble, reproduce its probability distribution.
  - Macroscopic properties can be estimated.
  - Applies to equilibrium molecular dynamics.

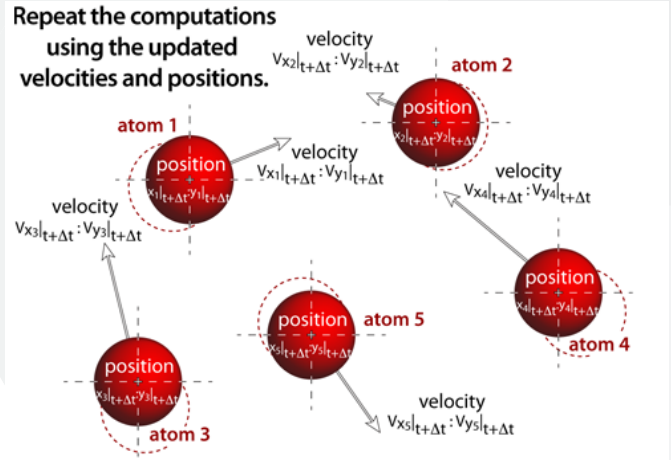


$$\langle A \rangle_{time} = \frac{1}{N} \sum_{i=1}^{N} A(t_i)$$

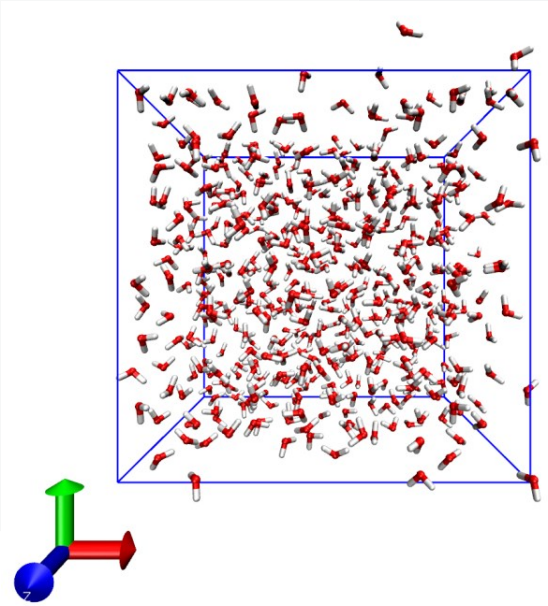
under equilibrium

Micro and macro

- If we have velocities, we can calculate kinetic energy and further, e.g., temperature and pressure.

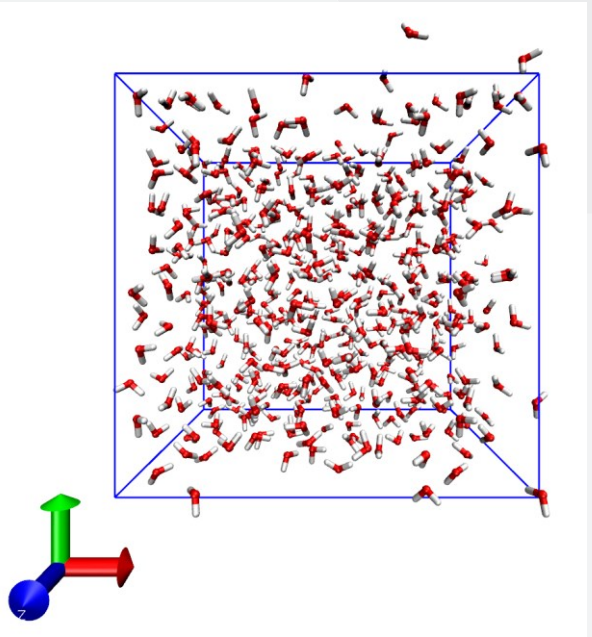


- Relation to pressure:
  - $p = \frac{2N}{3V}\left\langle \frac{1}{2}mv^2 \right\rangle$
- Relation to temperature:
  - $T = \frac{2}{3k_b}\left\langle \frac{1}{2}mv^2 \right\rangle$, k_b – Boltzmann constant



- However, Newton's equations of motion *per se* correspond to the constant energy in the system.
  - Temperature and pressure do not change unless they are adjusted by special algorithms .

Statistical ensembles in MD



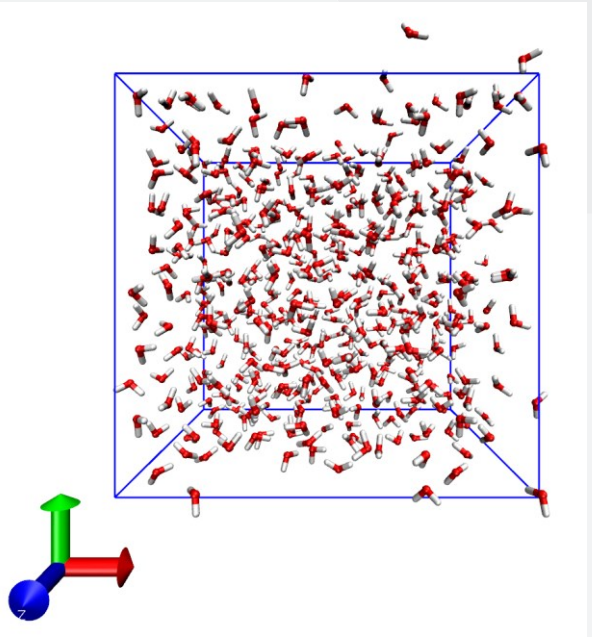
- Typically named by three letters.

- **NVE (microcanonical ensemble):**
  - N – number of particles (constant in a simulation).
  - V – volume (constant in a simulation).
  - E – energy (constant in a simulation).

- Energy is not added nor removed, fixed volume, particles just move in a locked box.
  - Newton's equations correspond to NVE (no external work, constant total energy, no heat exchange).

- Not used in practice: biological simulations are not isolated, big temperature fluctuations, not flexible, convergence and stability issues, etc.

Statistical ensembles in MD



- **NVT (canonical ensemble, isothermal, Helmholtz canonical):**
  - N – number of particles (constant in a simulation).
  - V – volume (constant in a simulation).
  - T – temperature (constant in a simulation).
- Temperature is maintained at a desired value through the exchange of energy with the surroundings (conceptually, of course).
- Sometimes used in bio: simple, stable, realistic enough, excellent for first equilibration stages.
- Disadvantages – volume needs to change in many biosystems, compressibility pressure fluctuations, etc.

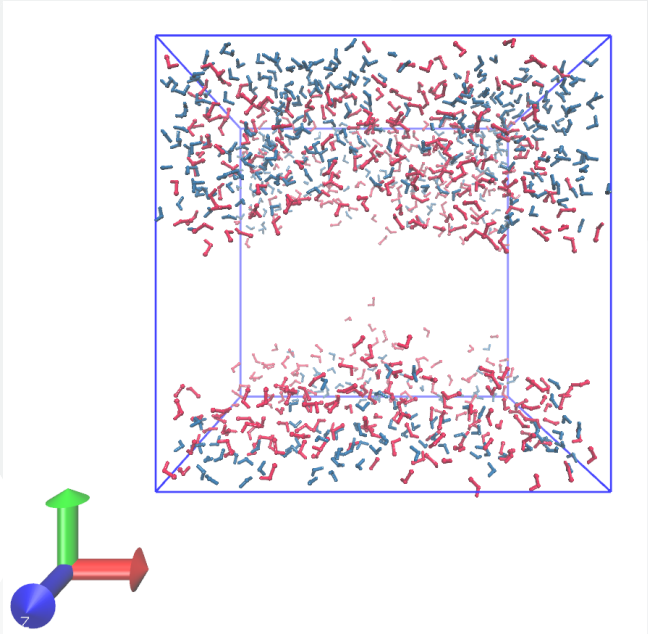
Statistical ensembles in MD

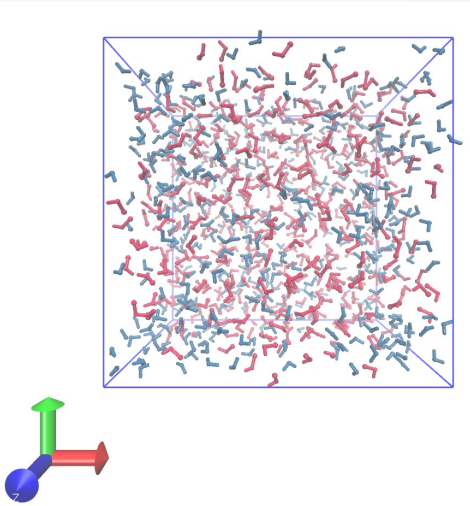
- **NPT (Gibbs canonical ensemble, isobaric-isothermal):**
  - N – number of particles (constant in a simulation).
  - P – pressure (constant in a simulation).
  - T – temperature (constant in a simulation).

- Both temperature and pressure are controlled; pressure is controlled through changing box dimensions.

- 99% use-case in biosimulations, replicates biological conditions, stable, flexible, direct comparison to experimental conditions.

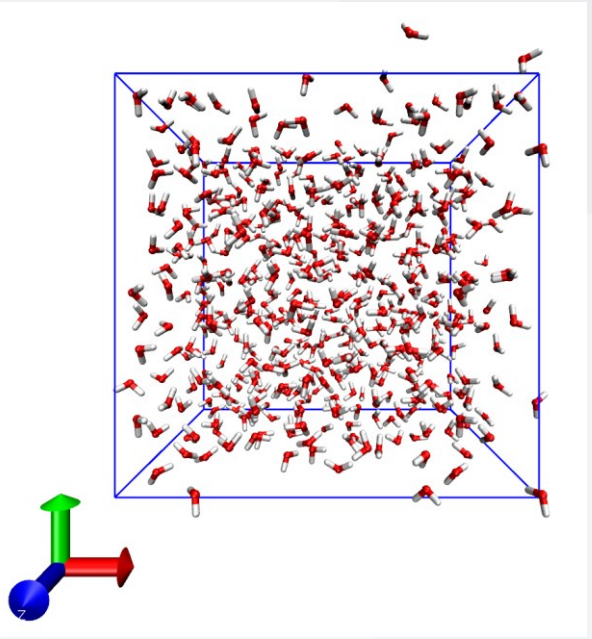
- Disadvantages – basically, technical and computational.





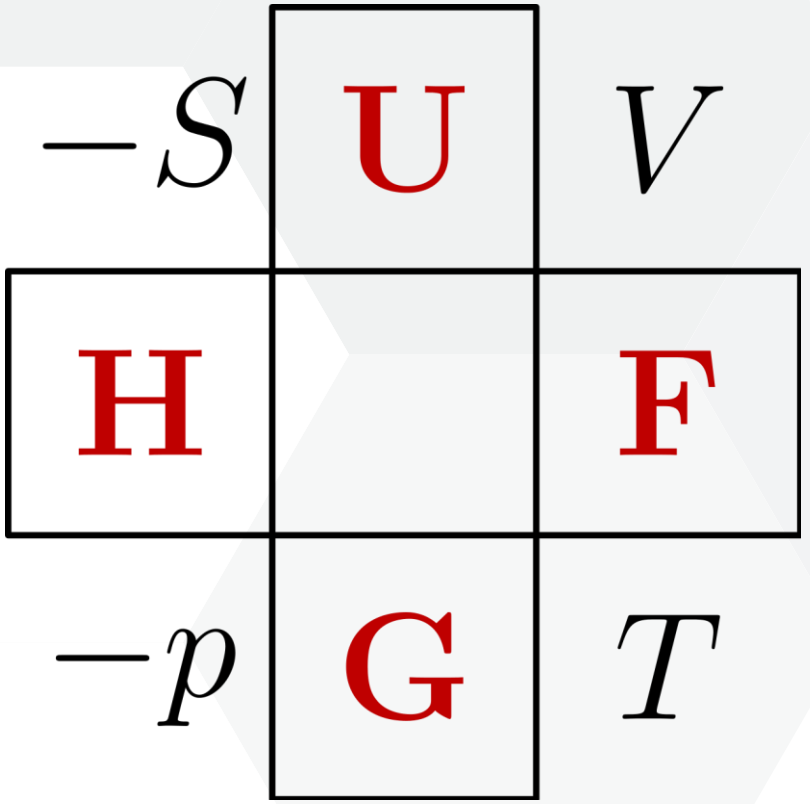
Statistical ensembles in MD

- Other examples:
  - NPH (isoenthalpic-isobaric: number of particles, pressure, enthalpy).
  - NPE (isobaric: number of particles, pressure, energy).
  - μVT (Grand canonical: chemical potential, volume, temperature).
  - μPT (chemical potential, pressure, temperature).
    - μVT and μPT are usually used in Monte-Carlo simulations.
  - NPγT (number of particles, pressure, surface tension, temperature).
  - …
- I think you will never use them.



Maxwell relations

- A set of equations in thermodynamics.
- Derived from the symmetry of second derivatives (complicated, I know).
- In simple words, these are equations that show how changes in one property (for instance, temperature) may affect another one (for instance, volume) under certain conditions.



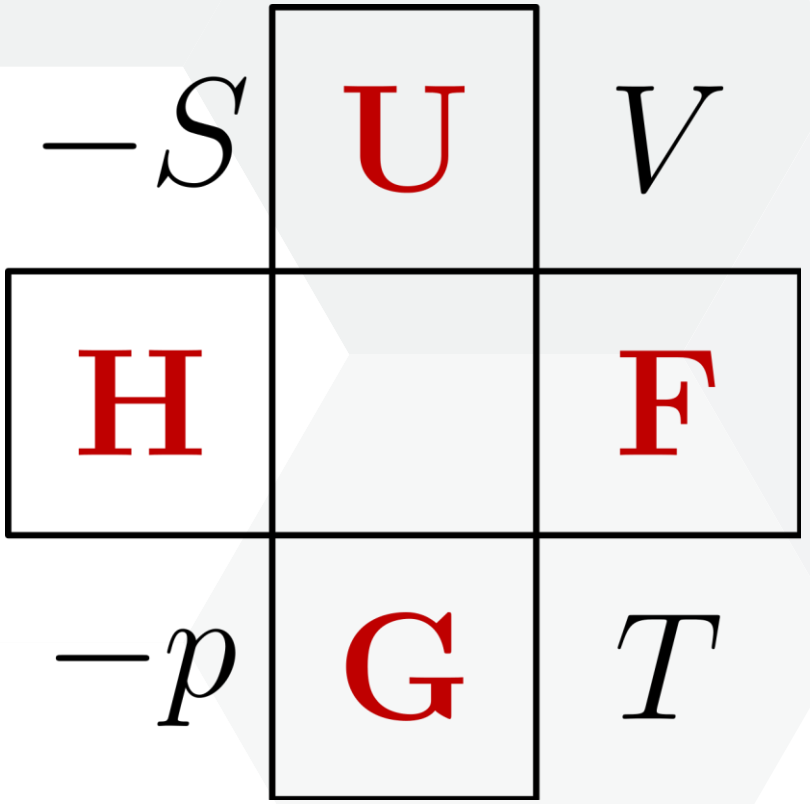
Thermodynamic square
picture from Wikipedia

Maxwell relations

- p – pressure
- V – volume
- T – temperature

- S – entropy

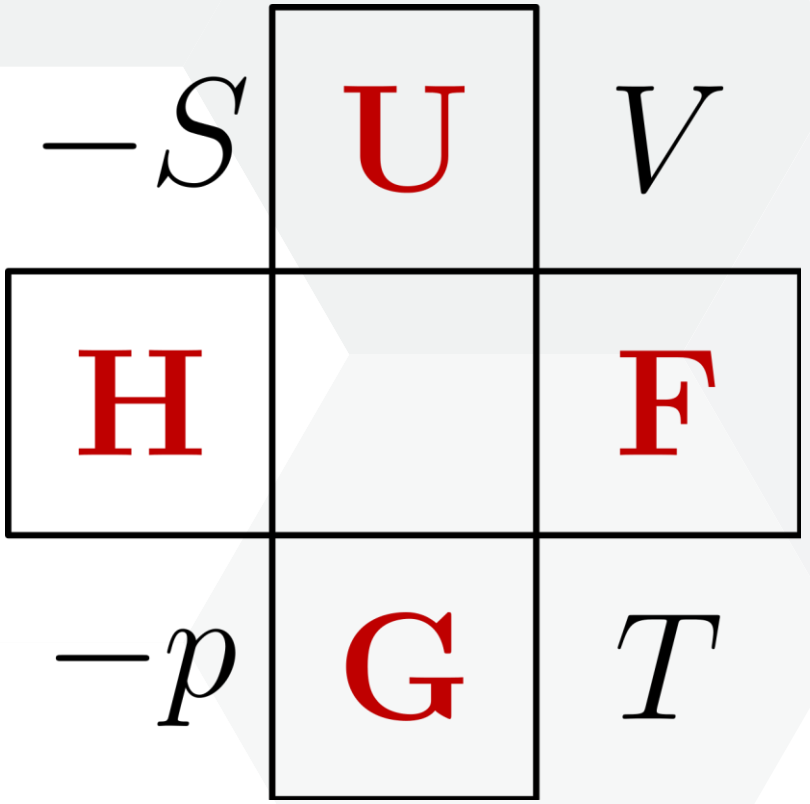
- U – internal energy
- H – enthalpy
- G – Gibbs free energy
- F – Helmholtz free energy (sometimes A)



Thermodynamic square
picture from Wikipedia

Maxwell relations

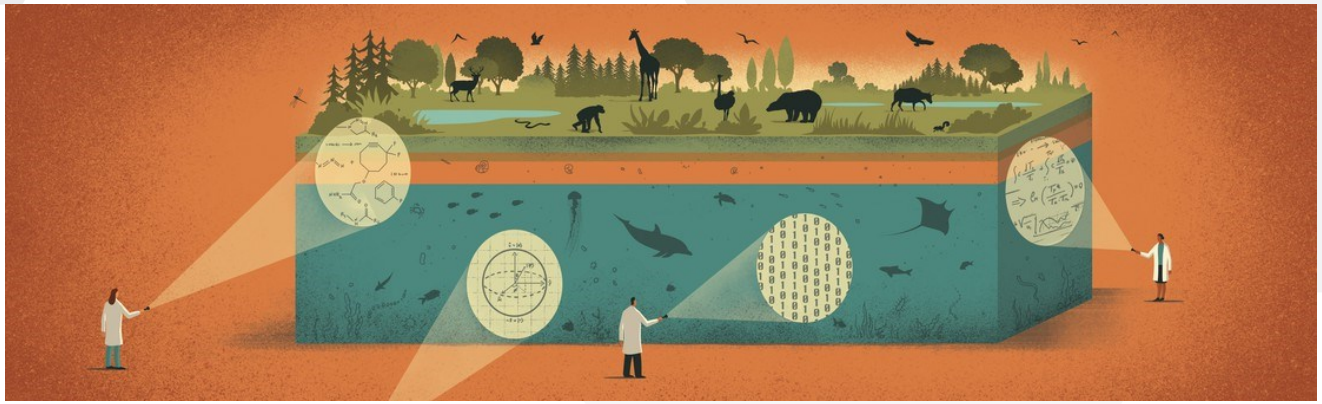
- U – internal energy
  - the total energy within a system
- H – enthalpy
  - energy of a system plus adsorbed/released heat at constant pressure
- G – Gibbs free energy
  - energy available to do work at constant temperature and pressure
- F – Helmholtz free energy
  - energy available to do work at constant temperature and volume



Thermodynamic square
picture from Wikipedia

Simulations coupled to experiments

- The beauty of simulations is when they are linked to experiments.
- Compare your results to experiments whenever possible.
  - Synergy, not competition.
- We are scientists, not film makers.
  - Produce science, not sci-fi.
- “Garbage in, garbage out”.
- “Good in, good out”?
  - More delicate concept to think about.



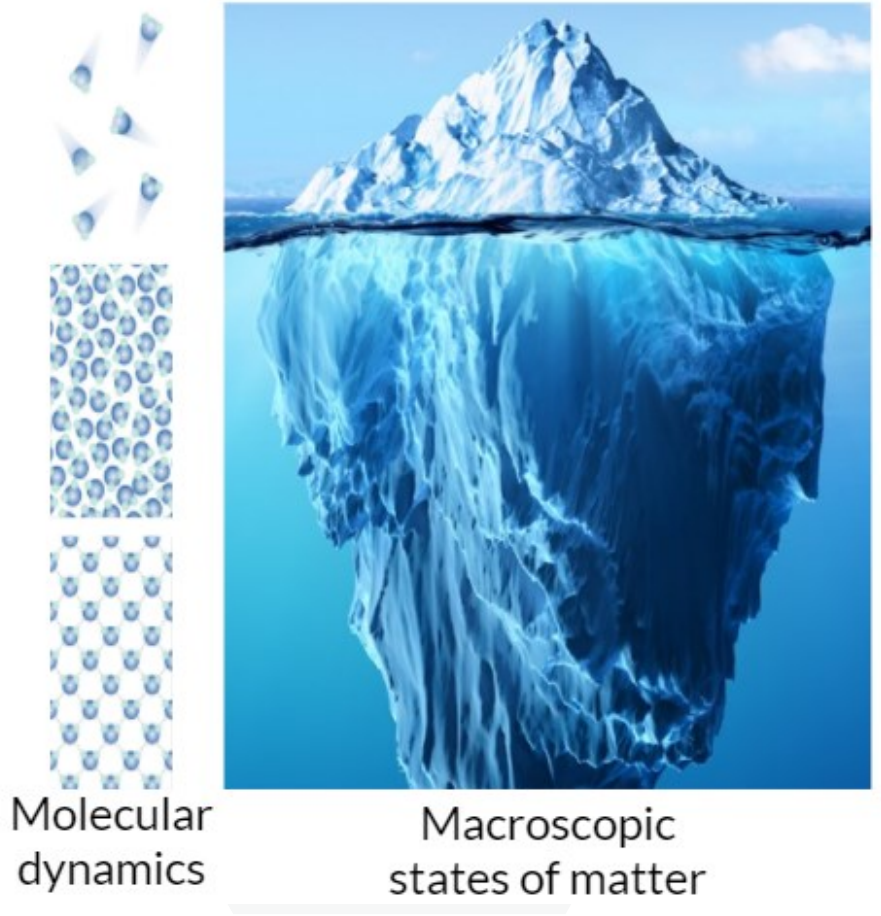
from https://elifesciences.org/subjects/physics-living-systems

Properties to calculate and analyze

- Macroscopic properties we already discussed:
  - Temperature, pressure, volume, energy (basically, must-do)
  - Surface tension, polarization, etc.

- Microscopic properties:
  - Atomic and molecular distribution via radial distribution functions
  - Ordering – orientation, dipoles, …
  - Conformations – secondary structure of proteins, biological assemblies, …
  - Interactions – hydrogen bonds, adsorption, ligand binding, selectivity, …
  - Motion – diffusion, translocation, opening/closing of molecular gates, …

- Collected quantities can be categorized, for instance, structural or dynamic (e.g., self-diffusion coefficient).
  - In today practical.



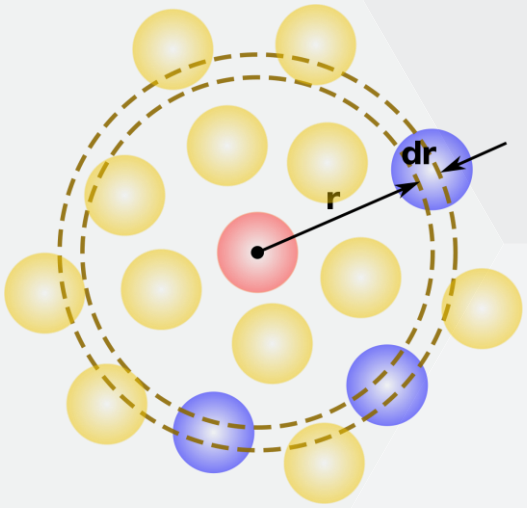
found on Twitter https://twitter.com/TrangAnhEstelle

Radial distribution function (RDF, $g(r)$)

- A measure of the probability to find a particle at a distance r from another reference particle.

- One of the pillars in post-processing MD simulations.
  - The easiest way to get the first “feeling” of what is happening in your simulations.
  - `gmx rdf`

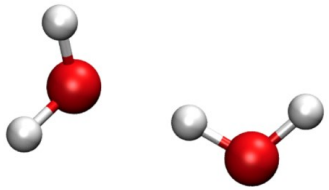
- RDF is related to the structure factor (via Fourier transformation), which can be determined from X-ray or neutron diffraction experiments.
  - One to one comparison of simulations and experiments.



picture from Wikipedia

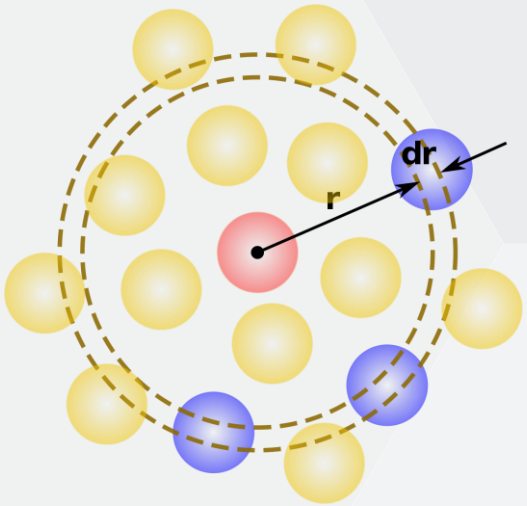
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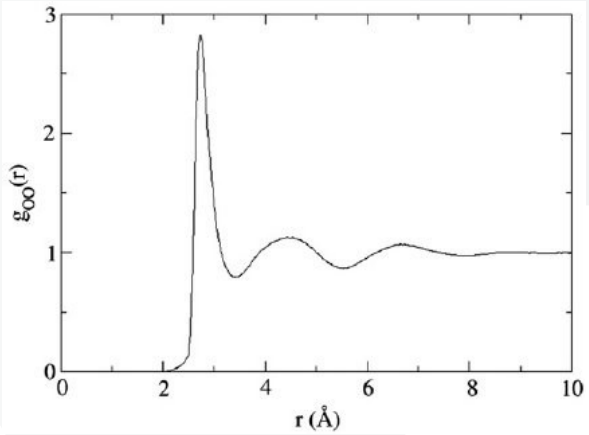


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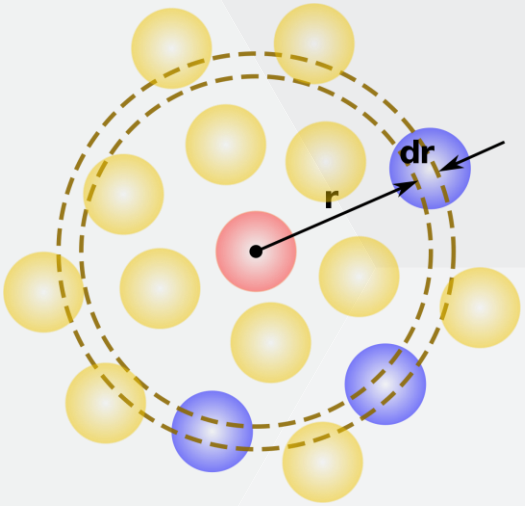


RDF of water oxygens around water oxygens from X-ray scattering data

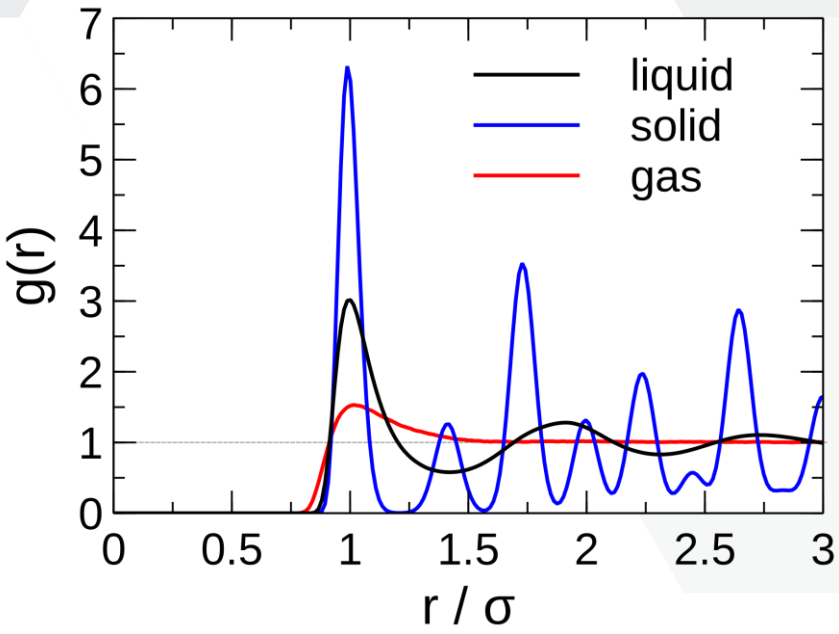
Clark et al. *Mol. Phys.* 2010, 108, 1415–1433

Radial distribution function (RDF, $g(r)$)

- Defined peaks? More ordering!
- Larger peaks? More ordering!
  - Do not confuse height, width, and area.
- In most cases (especially biosimulations), RDFs converges to the value of 1.
  - Value that does not equal to one means that there is correlation
  - Correlations are typically gone at long distances (few nanometers max).



picture from Wikipedia

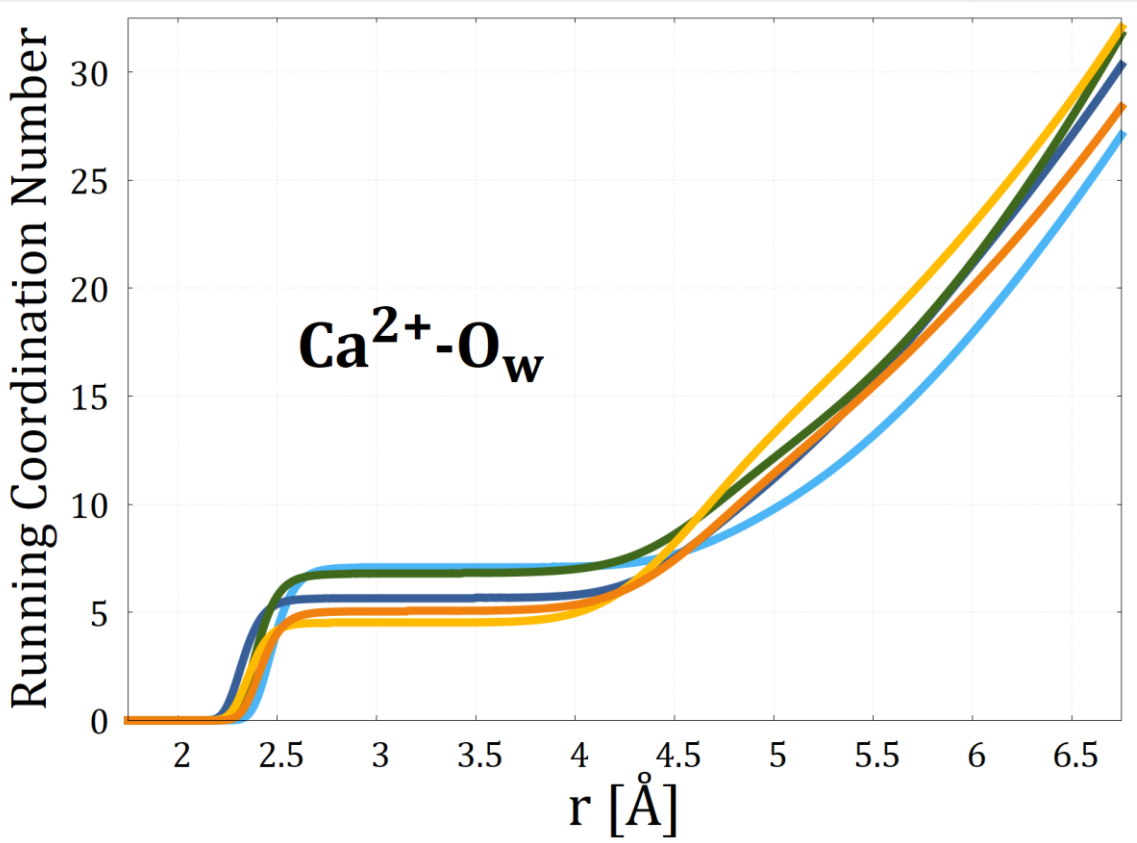


The radial distribution functions of solid (T = 50 K), liquid (T = 80 K), and gaseous argon (T = 300 K)

taken from Wikipedia

Running coordination number

- Coordination number is essentially the running integral of RDF.
  - You know what an integral is, right?
- Quick way to estimate how many atoms/molecules are around reference particle at a certain distance.
  - Useful for defined structures such a hydration shell of an ion, ligand, could be even protein.

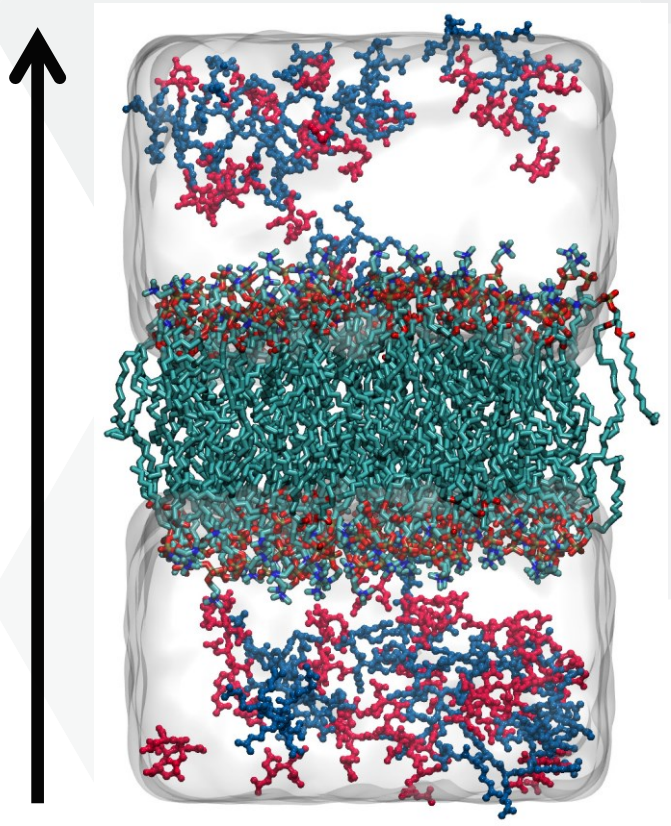


Running coordination number of water oxygens around calcium cation

Biriukov et al. *J. Chem. Phys.* 2022, 156, 194505

Density distribution (number/mass/charge density)

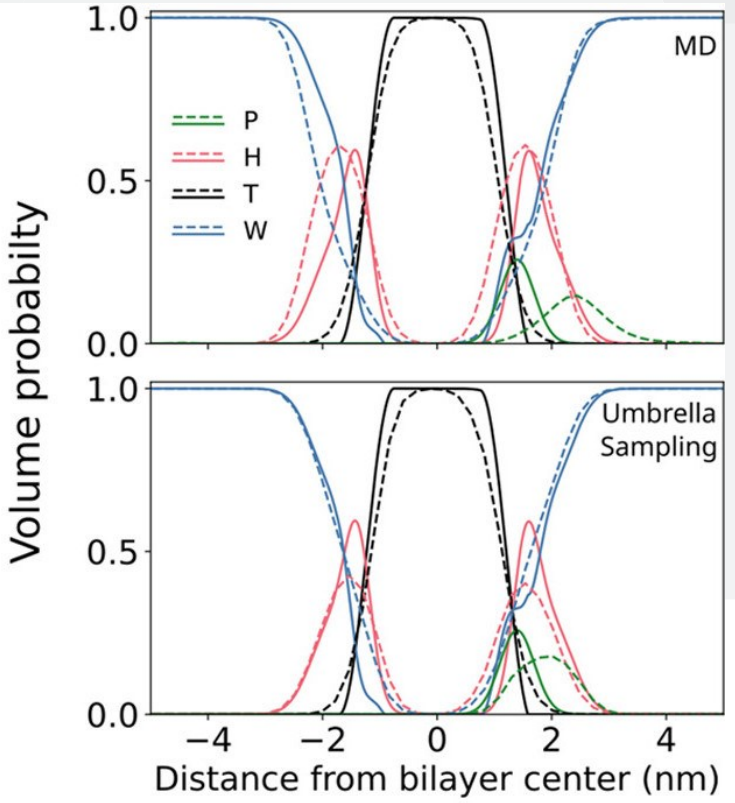
- Similar to RDF: distribution of particles but typically as a function of box dimension(s).
  - One-dimensional, lateral, three-dimensional, …
- Another pillar in post-processing MD simulations.
  - Especially relevant for interfacial systems. (membrane/aqueous interface, solid materials, water/air interface, etc.)
  - Inherently can possess information about orientation.
  - `gmx density`



Example of lipid bilayer system: density along the arrow would give you a lot of info

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  - `gmx density`
- Density can be also extracted from X-ray techniques and compared to simulations.
  - SAXS – small-angle X-ray scattering (see right).

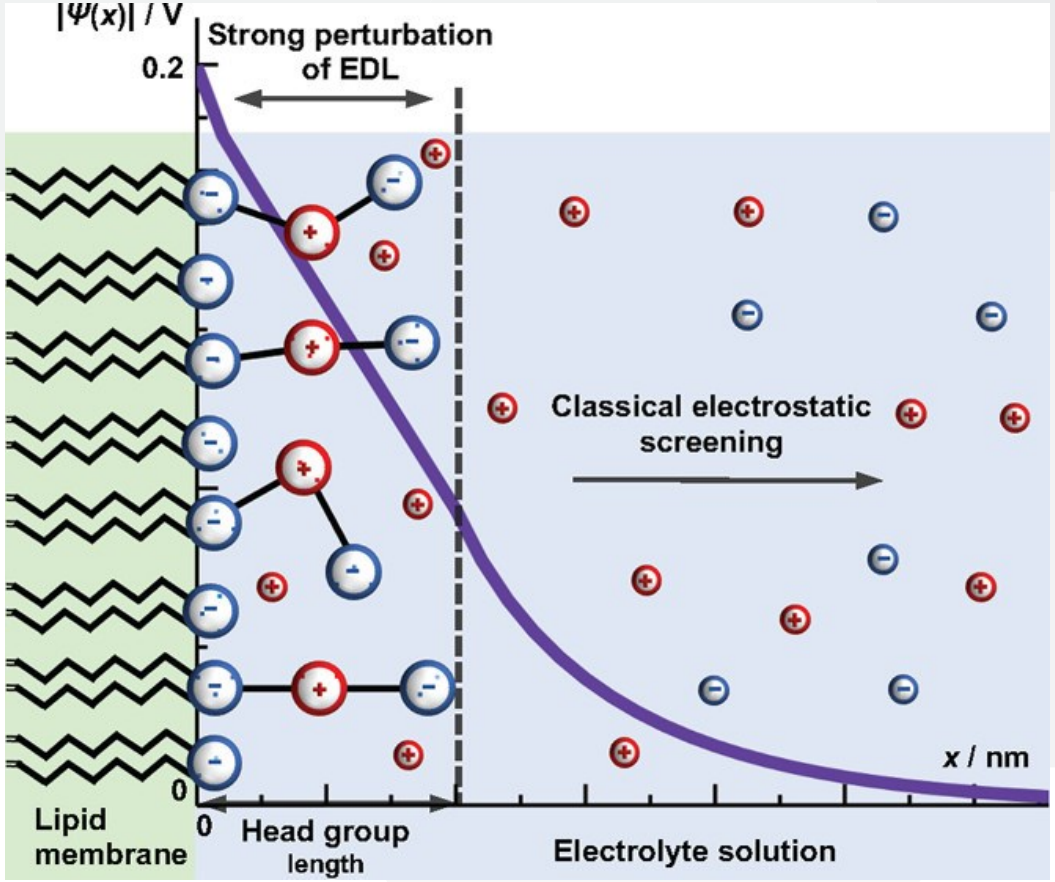


Density profile from simulations (dashed line) and SAXS experiments (solid line)

Carrer et al. *J. Phys. Chem. Lett.* 2023, 14, 7014–7019

Angular distributions

- Information that is not that easily accessible by experiments.
  - Good for people doing simulations!
- Simple to define, various information to obtain.
  - Protein/peptide/drug orientation when adsorbed to the membrane or translocated through it.
  - Response of lipid headgroups to these processes (see right).
  - `gmx angle`

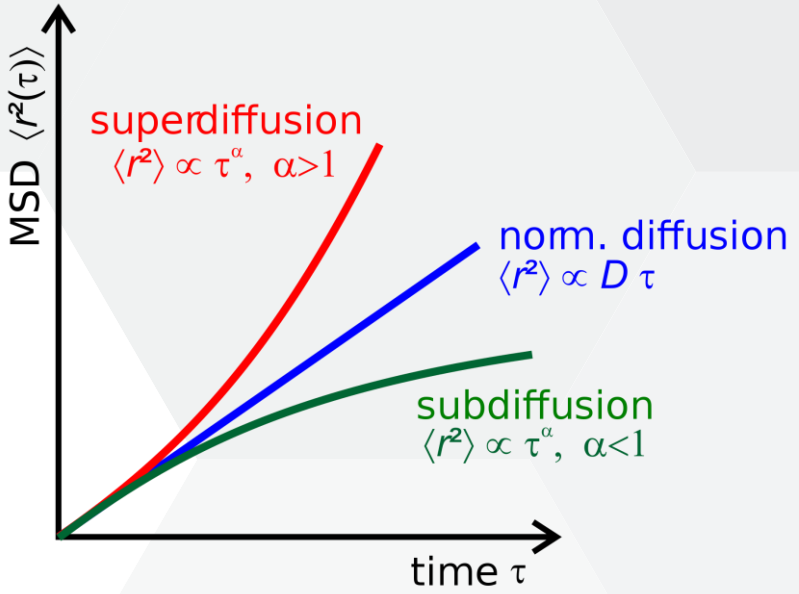


Schematic representation of lipid membrane/aqueous interface

Bohinc et al. *J. Chem. Theory Comput.* 2021, 18, 448–460

Self-diffusion coefficient

- Self-diffusion coefficient:
  - A transport property.
  - Quantifies how rapidly particles move in a medium.
  - Indicator of molecular mobility.
  - Related to viscosity, conductivity, structural properties, etc.
- Typically derived from the Einstein relation.
  - There are details though (see right, more during practical).
  - `gmx msd`
- Important fact: sensitive to the box size in your simulations.
  - Is it true? We will check it today.
  - Why is it happening? We will learn next week.



Comparison of normal and anomalous diffusions

taken from Wikipedia

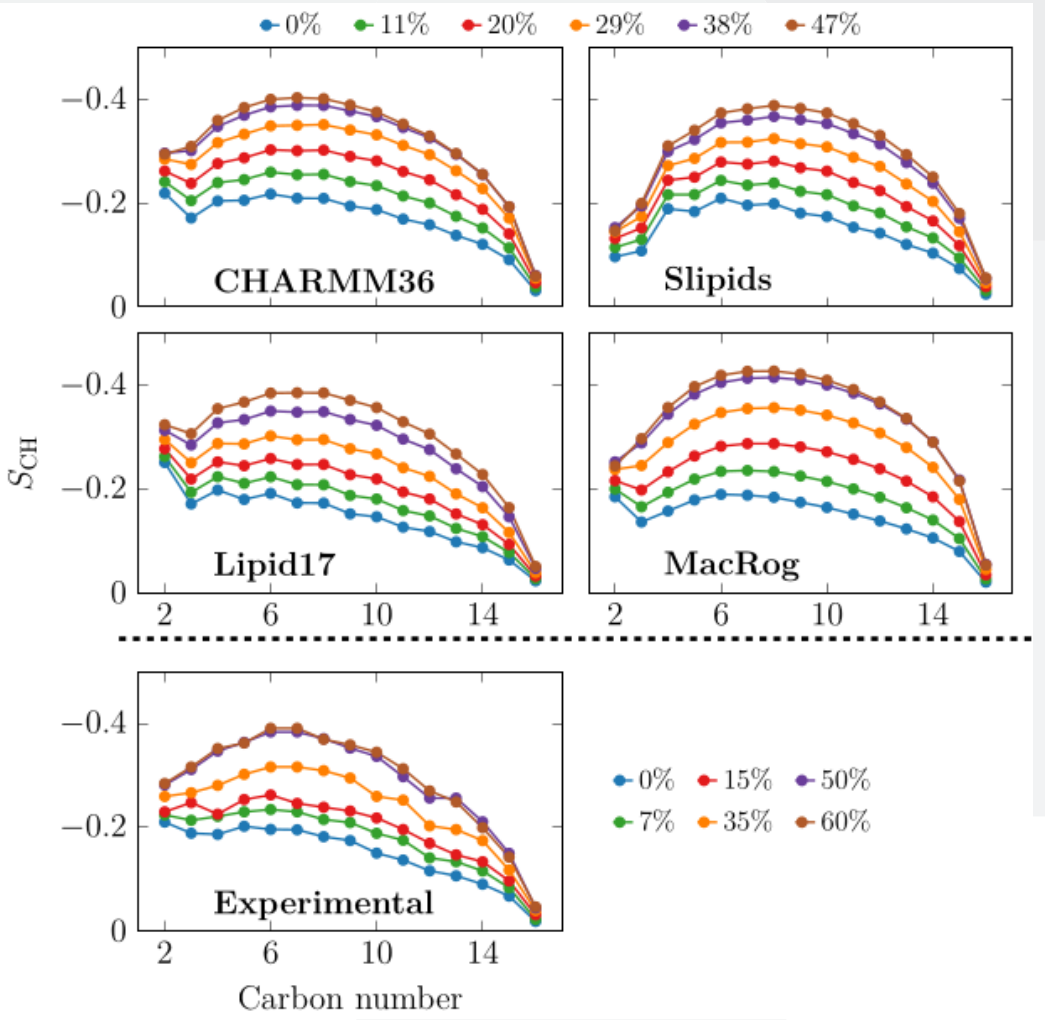
Other measures (not today)

- Order parameters of lipid membranes.
  - Alignment of lipid tails (orientation of C–H bond) with respect to the normal (angle θ).
  - Represents the ordering and packing of lipids.
  - Can be directly measured by both NMR and molecular simulations.
  - Highly relevant to describe multiple biological processes in cellular membranes.

- Commonly calculated as the deuterium order parameter.
  - $S = \langle 3\cos^2\theta - 1 \rangle / 2$
  - `gmx order`
  - Week 05.



Ondřej Kroutil



Order parameters for different lipid membranes with varying amount of cholesterol

Javanainen et al. *J. Chem. Theory Comput.* 2023, 19, 6342–6352

Other measures (not today)

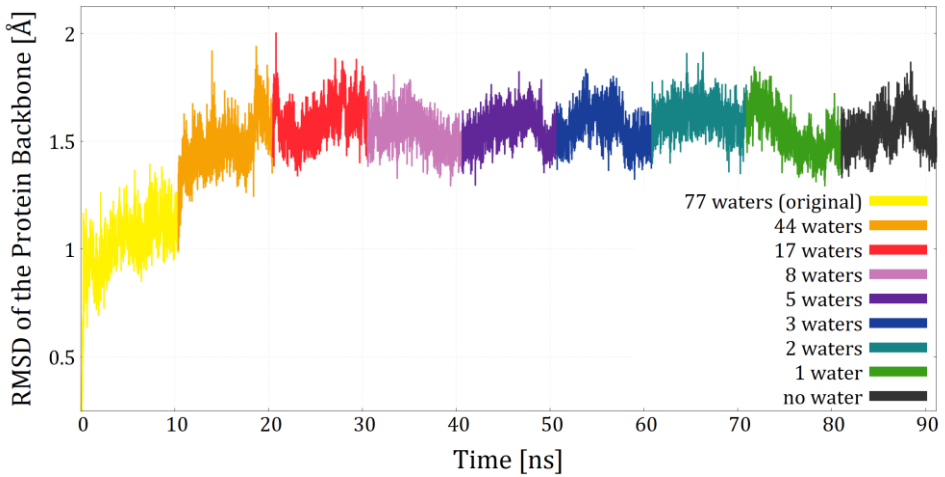
- Root-mean-square deviation (RMSD).
  - Must-do in simulations of proteins.
  - Excellent for equilibration monitoring and checking the stability of a protein structure.
  - `gmx rms`



Alignment of two proteins with a difference in a single mutation

taken from CHARMM-GUI

- Mutations/ligand binding can and usually do affect the RMSD of a protein.
  - If nothing is happening in your system, RMSD should become stable.



a typical RMSD graph

- `gmx dssp` – another tool to determine the secondary structure .

- Week 09.



Willian Morton

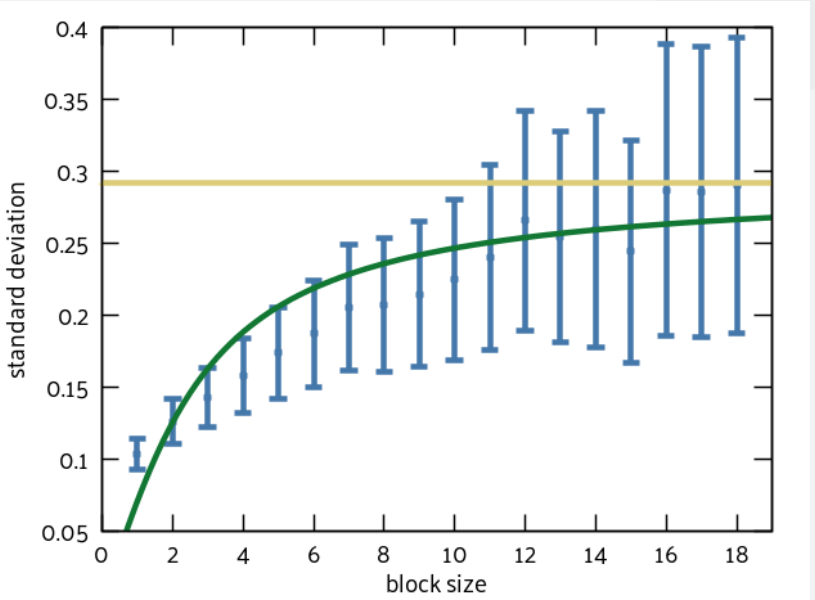
Kontkanen et al. *J. Chem. Phys.* 2022, 156, 175101

Error analysis

- Whatever you report or even if you think your simulation is fantastic – error estimates are the must.

- Simulations are well-suited for reporting errors.
  - We calculate averages.
  - Time correlations.
  - Error of a mean.

- Standard deviation/error from:
  - Replica simulations = multiple simulations.
  - Divide one simulation in pieces.
    - In principle, good, but not ideal.
  - Block averaging.
    - `gmx analyze`



Example of block averaging: standard deviation depends on the size of the block

https://realerthinks.com/block-averaging-bootstrapping-estimating-mean-autocorrelated-data/

Break!

Get ready for practical session!